8.1 Analytical Techniques (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	8. Analysis (A Level Only)
Topic	8.1 Analytical Techniques (A Level Only)
Difficulty	Easy

Time allowed: 50

Score: /39

Percentage: /100

Question 1a

This question is about gas chromatography.

Name the **three** types of chemical that gas chromatography is used for.

[3 marks]

Question 1b

Silica and alumina are commonly used as the stationary phase in thin layer chromatography.

State the **type** of chemical that is commonly used as the stationary phase in gas chromatography.

[2 marks]

Question 1c

State the type of chemical that is used for the mobile phase in gas chromatography. You should include at least **one** specific example in your answer.

[2 marks]

Question 1d

Results in gas chromatography are based on retention time.

i)
Define the term retention time.

[1]

ii)
State **three** factors that retention time depends upon.

[3]

[4 marks]

Question 1e

State what information the relative size or area under the peak on a gas chromatogram provides.

[1 mark]

Question 1f

Gas chromatography is carried out using a polar stationary phase and argon as the carrier gas.

Use the chromatogram in Fig. 1.1 to answer the following questions.

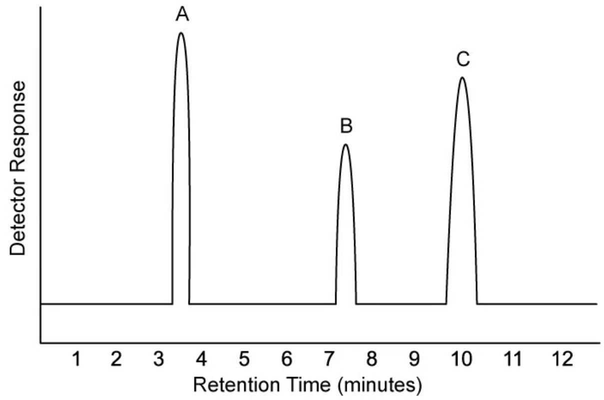


Fig. 1.1

i)
State which compound has the lowest retention time.

[1]

ii)
State which compound is the least polar.

[1]

iii)
State which compound has the greatest interaction with the stationary phase.

[1]

[3 marks]

Question 2a

This question is about the NMR analysis of various organic compounds.

Name and draw the structure of the chemical that is commonly used as a standard in NMR spectroscopy.

[2 marks]

Question 2b

Fig. 2.1 shows the structures of compounds **A**, **B** and **C**.

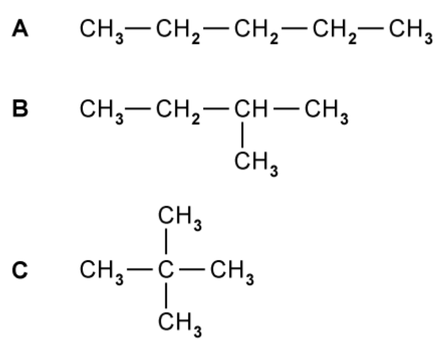


Fig. 2.1

Compound **A** is pentane, with the chemical formula C_5H_{12}. Compound **B** is 2–methylbutane and compound **C** is 2,2–dimethylpropane, which are both isomers of pentane.

State the number of hydrogen peaks that would be expected in low resolution ^{1}H-NMR spectrum of each isomer.

[3 marks]

Question 2c

More structural details can be deduced using high resolution ^1H NMR.

Explain why the methyl groups in 2–methylbutane, compound **B**, give a doublet splitting pattern while the methyl groups in 2,2–dimethylpropane, compound **C**, give a singlet splitting pattern.

[3 marks]

Question 2d

Carbon-13 NMR is also commonly used to distinguish chemicals.

Predict the number of peaks in the carbon-13 NMR spectra of compounds **A**, **B** and **C**.

[3 marks]

Question 3a

This question is about thin layer chromatography.

Name the **two** phases of chromatography and give **one** example of a chemical used for each phase.

[4 marks]

Question 3b

Thin layer chromatography is used to check if an unknown food colouring contains a banned colouring.

Draw a labelled diagram to show the experimental setup for this TLC analysis.

[3 marks]

Question 3c

State two factors that the rate of separation depends upon.

[2 marks]

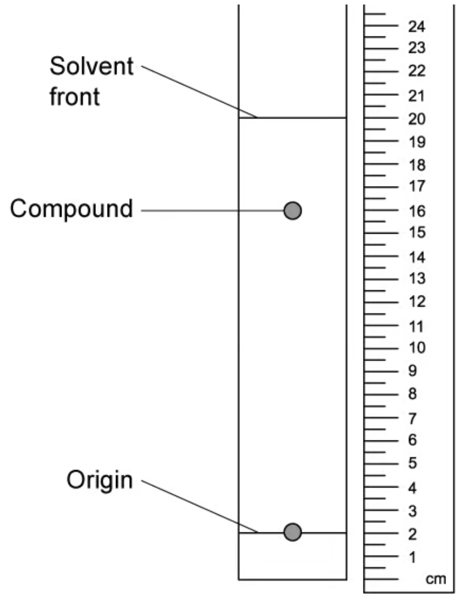
Question 3d

State the equation used to calculate the unique retention factor of a compound.

[1 mark]

Question 3e

Calculate the R_f value of the compound shown in the chromatogram below.



[3 marks]

Model Answers: Easy

1a

a) The **three** types of chemical that gas chromatography is used for are:

- Gases; [1 mark]
- Volatile liquids; [1 mark]
- Solids in their vapour form; [1 mark]

[Total: 3 marks]

- Gas chromatography (also called gas-liquid chromatography) is used for identifying and separating chemicals that cannot be tested for by thin-layer chromatography
- If you continue your chemistry studies, you will learn that there are more types of chromatography
- If you think about the answers, they are all linked to gases which would be hard to spot on a TLC plate, unless you could dissolve them into a solvent without losing or reacting them

1b

b) The **type** of chemical commonly used as the stationary phase in gas-liquid chromatography is a:

- A non-polar liquid (on a solid support); [1 mark]
- Which is non-volatile
  OR
  Has a high boiling point; [1 mark]

[Total: 2 marks]

- **Remember:** Volatile means that the liquid can easily change state to become a gas

    - This is why a non-volatile liquid / a liquid with a high boiling point is used to be used as the stationary phase

1c

c) The mobile phase in gas chromatography is:

- An inert (carrier) gas; [1 mark]

Any **one** from the following:

- Helium; [1 mark]
- Nitrogen; [1 mark]
- Argon; [1 mark]

[Total: 2 marks]

- The alternative name for gas chromatography is gas-liquid chromatography
    - This gives a clearer indication of the phases involved
    - One phase is gaseous and the other is liquid
- The mobile phase is gaseous and therefore must be inert so that there are no reactions with any of the components being separated

1d

d)

i) Retention time is:

- The time taken for a component to travel through the column
  OR
  The time taken from injection to detection of the sample; [1 mark]

ii) The retention time depends upon:

Any **three** of the following:

- The attraction between the sample and the stationary phase; [1 mark]
- The attraction between the sample / component and the mobile phase; [1 mark]
- The polarity of the sample / component; [1 mark]
- The volatility of the sample / component; [1 mark]

- The nature / chemical structure of the sample / component; [1 mark]

[Total: 4 marks]

- If you think logically about the phrase retention time, you should be able to come up with the definition
  - It is the **time** that a component / sample is **retained**
  - Now, you just have to focus on where the component / sample is retained
  - It must be retained in the column
- Gas chromatography is no different to any other chromatography in terms of depending on the interactions between the sample / component and the mobile and stationary phases
  - These interactions themselves depend on the polarity, volatility and nature / chemical structure of the sample / component

1e

e) The relative size or area under the peak on a gas chromatogram tells you:

- The relative concentration / amount of each component; [1 mark]

[Total: 1 mark]

- The peaks of a gas chromatogram are similar to a Maxwell-Boltzmann curve
- The height / area under the peak gives you more information about that specific peak
- In gas-liquid chromatography, this information is the relative amount of that specific compound / component

1f

f)

i) The compound with the lowest retention time is:

- **A**; [1 mark]

ii) The least polar compound is:

- **A**; [1 mark]

iii) The compound that has the greatest interaction with the stationary phase is:

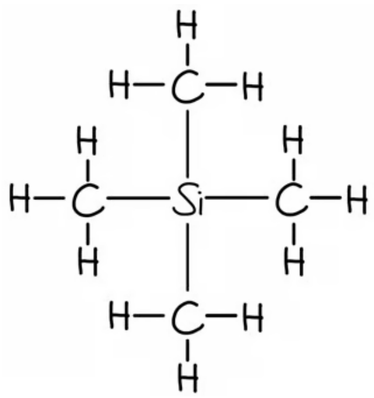
- **C**; [1 mark]

[Total: 3 marks]

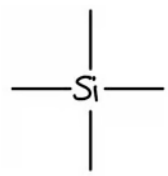
- For part **(i)**, you just read off the graph
- For part **(ii)**, you must recognise that the least polar compound will be the one with the lowest retention time when the stationary phase is polar
  - This is essentially another way of asking the question from part **(i)**
- For part **(iii)**, the compound that has the greatest interaction with the stationary phase is the one that travels the slowest and therefore has the highest retention time

2a

a) The name and structure of the chemical that is commonly used as a standard in NMR spectroscopy are:

- Tetramethylsilane; [1 mark]
- 

  OR

   ; [1 mark]

[Total: 2 marks]

- Whilst the abbreviation TMS is commonly used, it is not sufficient for the mark

- When drawing the structure, a partially displayed structure with CH_3 groups not displayed would be accepted
- You should be aware of TMS as the common reference standard for both ^{13}C NMR and ^{1}H NMR spectroscopy
- You should be able to name it, draw it and explain why it is used

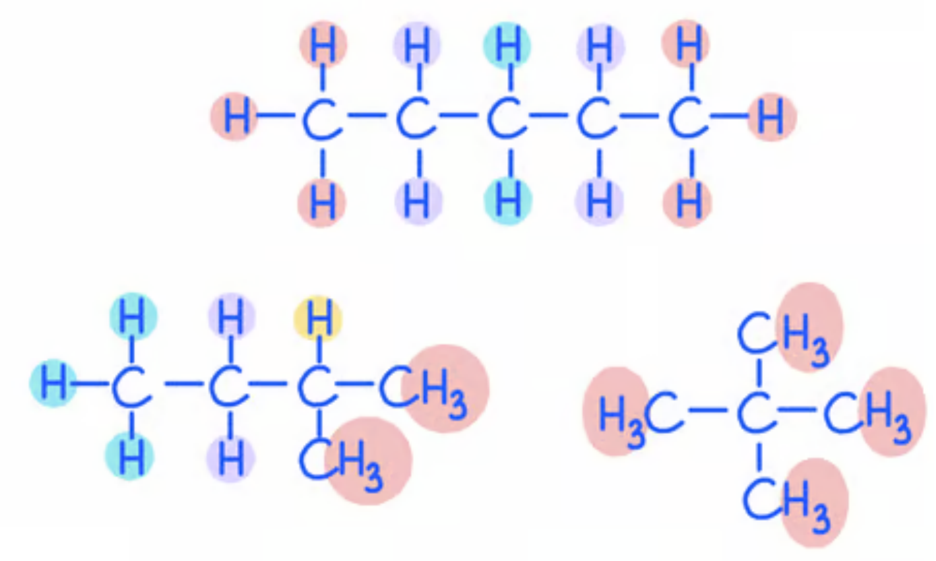
2b

b) The number of expected peaks in the ^{1}H-NMR of each compound is:

- **A** = 3; [1 mark]
- **B** = 4; [1 mark]
- **C** = 1; [1 mark]

[Total: 3 marks]

- **Tip:** Draw out and annotate the compounds as this can help you see the environments
  - You can sometimes make spotting different hydrogen environments easier by using symmetry - always double check your thinking though!
- The diagrams below show the different hydrogen environments



- Each peak on the NMR spectrum relates to hydrogens in the same environments, i.e. 3 different hydrogen environments would have 3 peaks

2c

c) The methyl groups in compounds **B** and **C** give a doublet and singlet respectively because:

- The splitting pattern depends on the neighbouring hydrogens/protons; [1 mark]
- Isomer **B** gives a doublet as the methyl groups have 1 neighbouring proton
  AND
  n + 1 = 1 + 1 = 2; [1 mark]
- Isomer **C** gives a singlet as the methyl groups have no neighbouring protons
  AND
  n + 1 = 0 + 1 = 1; [1 mark]

[Total: 3 marks]

- The splitting pattern of any hydrogen is dependent on its environment / the neighbours
- **Remember**: The number of peaks a signal splits into = n + 1
  - Where n = the number of hydrogens on the adjacent carbon
  - So if the methyl group has 1 neighbouring hydrogen, this will show as a doublet as 1 + 1 =2
- The same functional / alkyl groups may not have the same splitting patterns and chemical shifts

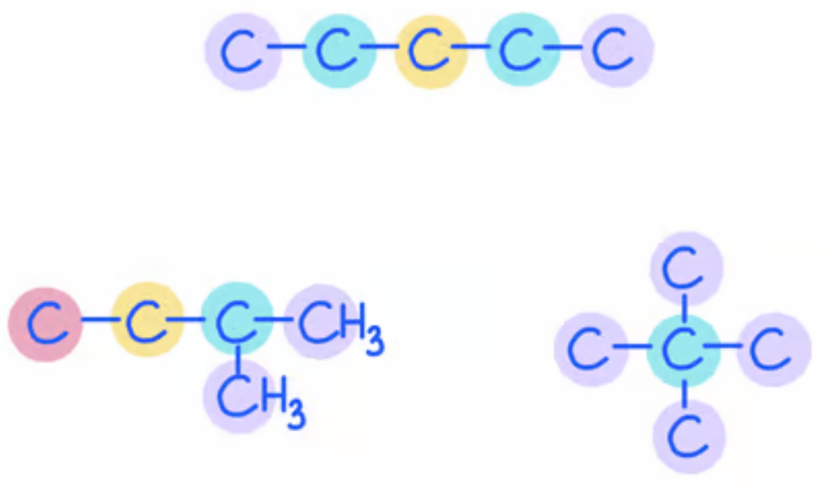
2d

d) The number of different carbon environments in each of the compounds is:

- **A** = 3; [1 mark]
- **B** = 4; [1 mark]
- **C** = 2; [1 mark]

[Total: 3 marks]

- **Tip:** You can sometimes make spotting different carbon environments easier by not drawing in all of the hydrogens as well as using symmetry - always double-check your answer
- In the diagram below the different highlights represent the different carbon environments
  - Make sure you **do** include hydrogens within functional groups, e.g. CHO, to make sure you aren't missing carbon environments that are actually different



3a

a) The **two** phases of chromatography and **one** example of a chemical used for each phase are:

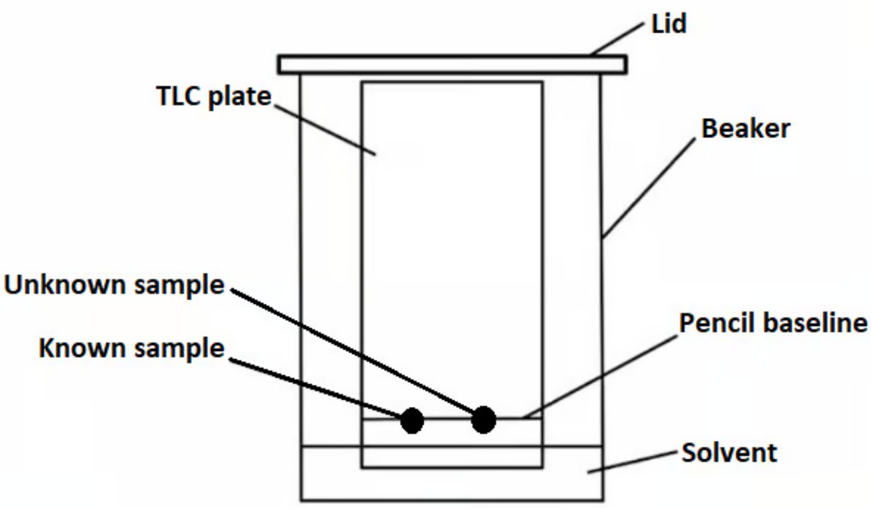
- The mobile phase; [1 mark]
- Specific named solvent, e.g. water / ethanol / hexane
  OR
  General solvent, e.g. alkane / alcohol; [1 mark]
- The stationary phase; [1 mark]
- Silica / silicon dioxide / SiO_2
  OR
  Alumina / aluminium oxide / Al_2O_3; [1 mark]

[Total: 4 marks]

- The two phases in chromatography are the stationary phase and the mobile phase
  - The stationary phase is the medium that the sample is travelling through, e.g. for TLC this is often silica
  - The mobile phase is the solvent and sample mixture that is being separated / purified
    - Common examples of polar solvents are water or alcohol
    - Common examples of non-polar solvents are the alkanes

3b

b) The experimental setup for this TLC analysis is:



- Pencil baseline labelled
  AND
  Lid on the beaker; [1 mark]
- Solvent labelled
  AND
  Level of solvent below pencil baseline; [1 mark]
- Two spots labelled known sample and unknown sample; [1 mark]

[Total: 3 marks]

- When drawing the equipment used in TLC, it is common that students omit the lid, but this is needed to prevent the evaporation of often volatile organic solvents
- **Careful:** Examiners will look very closely at your diagram to check that your solvent level isn't above the pencil baseline
  - There have been exams where having the solvent touching the spots of the sample has cost marks

3c

c) Two factors that the rate of separation depends upon are:

- The solubility of the sample / mixture / compounds / components in the mobile phase; [1 mark]

- The retention of the sample / mixture / compounds / components with the stationary phase
  OR
  How the sample / mixture / compounds / components interact with the stationary phase; [1 mark]

Level of solvent below pencil baseline; [1 mark]
- Two spots labelled known sample and unknown sample; [1 mark]

[Total: 3 marks]

- When drawing the equipment used in TLC, it is common that students omit the lid, but this is needed to prevent the evaporation of often volatile organic solvents
- **Careful:** Examiners will look very closely at your diagram to check that your solvent level isn't above the pencil baseline
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3c

c) Two factors that the rate of separation depends upon are:

- The solubility of the sample / mixture / compounds / components in the mobile phase; [1 mark]
- The retention of the sample / mixture / compounds / components with the stationary phase
  OR
  How the sample / mixture / compounds / components interact with the stationary phase; [1 mark]

[Total: 2 marks]

- The rate of separation must depend on the two components of chromatography; the mobile and the stationary phases
  - For the mobile phase, it is about how soluble the compounds are
  - For the stationary phase, it is about the strength of the interactions between the sample and the stationary phase

3d

d) The equation to calculate the unique retention factor of a compound is:

- $R_f = \frac{\text{distance travelled by component}}{\text{distance travelled by solvent}}$; [1 mark]

[Total: 1 mark]

- This equation isn't often specifically asked for in a question
- You are more likely to be asked to apply this equation to calculate an R_f value

3e